

Machine Learning

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

Problem Statement:

The aim of this project is to predict diabetes risk using patient medical attributes such as glucose, blood pressure, BMI, insulin, and age. Early detection helps doctors provide timely lifestyle guidance and medical support.

Case-Study Scenario: Diabetes Risk Prediction

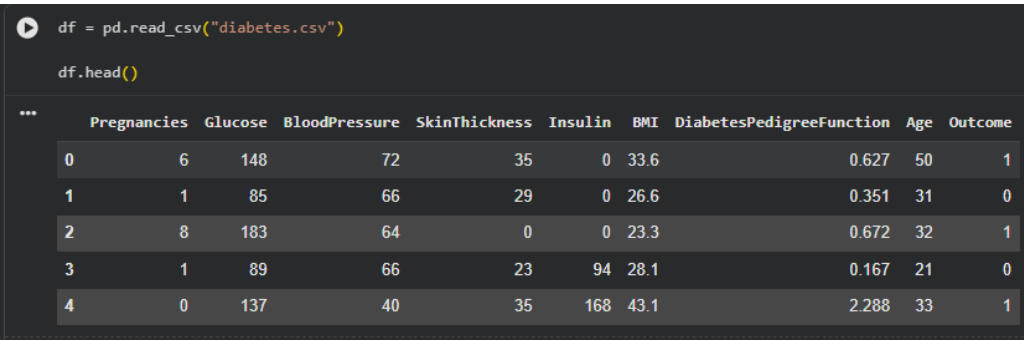
Imagine a community health centre that regularly collects medical data from patients, including glucose levels, blood pressure, BMI, insulin levels, and age. The doctors want to identify patients who are at high risk of developing diabetes so they can provide early lifestyle guidance and medical support.

- A patient named Anitha, aged 45, visits the clinic. Her glucose level is slightly elevated, BMI is high, and she has a family history of diabetes.
- The clinic feeds her health data into the machine learning system trained on the PIMA Indians Diabetes Dataset.
- The system runs two models:
 1. Logistic Regression gives a probability score, showing that Anitha has a 70% chance of being diabetic.
 2. Random Forest, with its ensemble of decision trees, confirms the risk with higher accuracy by analyzing complex feature interactions.
- To ensure reliability, the clinic uses k-fold cross validation, which validates the model's predictions across multiple data splits.

Based on these results, the doctors classify Anitha as “high risk” and recommend immediate lifestyle changes and further medical tests.

Exploratory Data Analysis (EDA):

Dataset Overview



```
df = pd.read_csv("diabetes.csv")
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

```

Dataset Shape:
(768, 9)
...
Column Names:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          768 non-null    int64
1   Glucose                              768 non-null    int64
2   BloodPressure                        768 non-null    int64
3   SkinThickness                        768 non-null    int64
4   Insulin                              768 non-null    int64
5   BMI                                  768 non-null    float64
6   DiabetesPedigreeFunction             768 non-null    float64
7   Age                                  768 non-null    int64
8   Outcome                              768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB

Statistical Summary:

```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

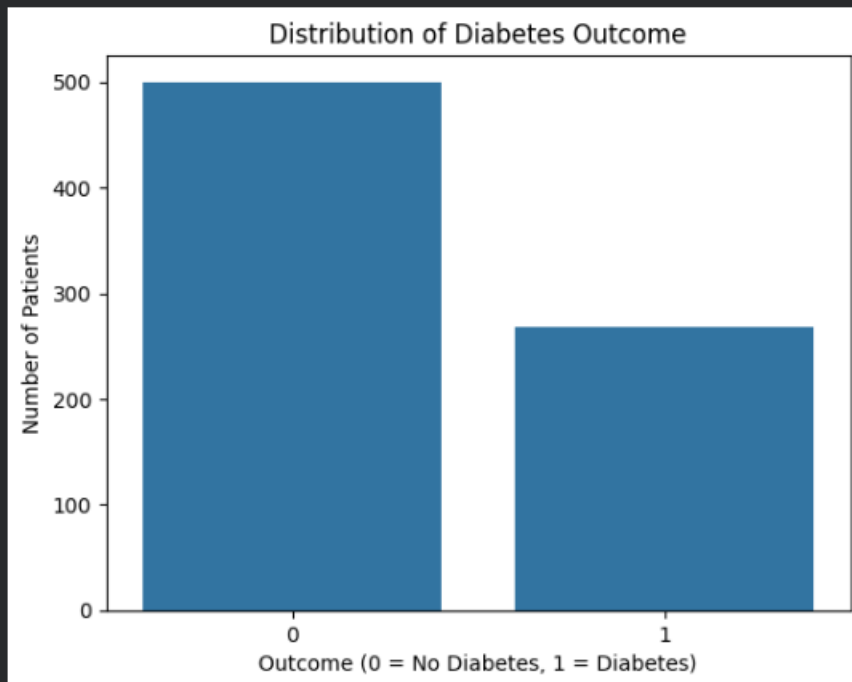
The dataset contains 768 records and 9 attributes related to medical measurements such as Glucose, Blood Pressure, BMI, Insulin, and Age. All columns have non-null values, but some features contain zero values that are medically unrealistic (e.g., Glucose = 0). This dataset is the PIMA Indians Diabetes Dataset, commonly used for diabetes prediction.

Outcome Distribution

```
sns.countplot(x='Outcome', data=df)

plt.title("Distribution of Diabetes Outcome")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Number of Patients")

plt.show()
```



The distribution shows that 65% of patients are non-diabetic (Outcome = 0) and 35% are diabetic (Outcome = 1).

This imbalance indicates that the dataset is slightly skewed toward non-diabetic cases.

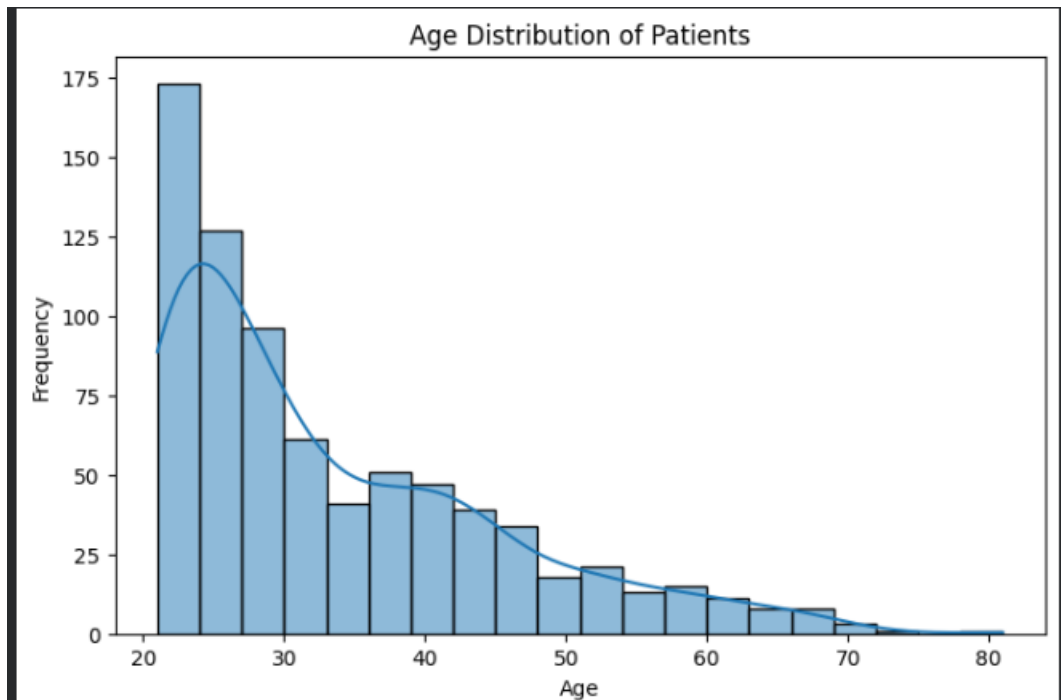
Age Distribution

```
plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=20, kde=True)

plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



Most patients are between 20 and 40 years old, with fewer cases in older age groups. The distribution is right-skewed, meaning younger individuals dominate the dataset.

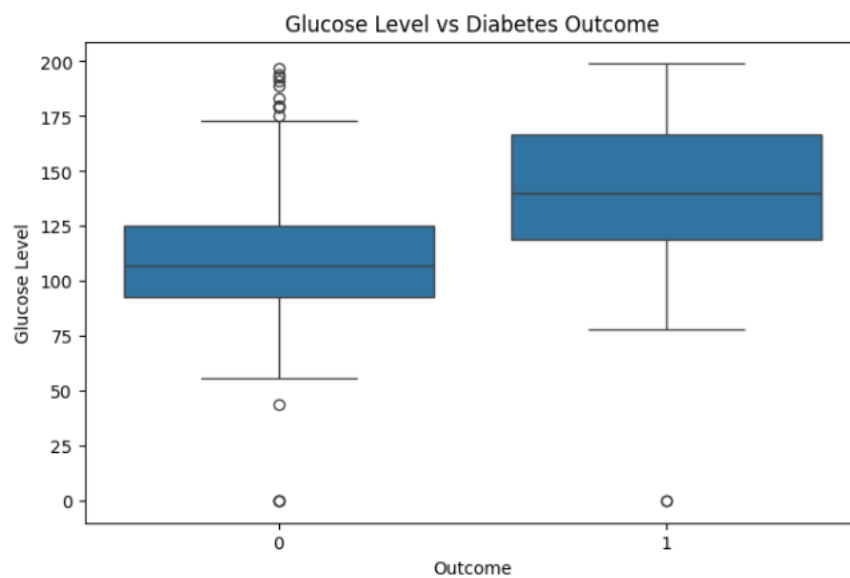
Glucose Level vs Diabetes Outcome

```
plt.figure(figsize=(8,5))

sns.boxplot(x='Outcome', y='Glucose', data=df)

plt.title("Glucose Level vs Diabetes Outcome")
plt.xlabel("Outcome")
plt.ylabel("Glucose Level")

plt.show()
```



Diabetic patients (Outcome = 1) have higher glucose levels compared to non-diabetic patients.

This confirms that glucose is a strong predictor of diabetes.

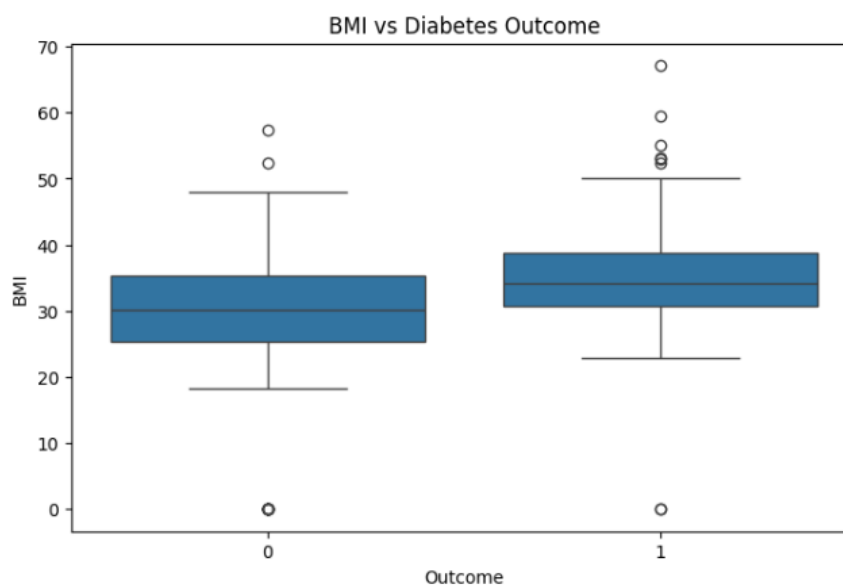
BMI vs Diabetes Outcome

```
plt.figure(figsize=(8,5))

sns.boxplot(x='Outcome', y='BMI', data=df)

plt.title("BMI vs Diabetes Outcome")
plt.xlabel("Outcome")
plt.ylabel("BMI")

plt.show()
```



Diabetic patients tend to have higher BMI values, showing that obesity is an important risk factor for diabetes.

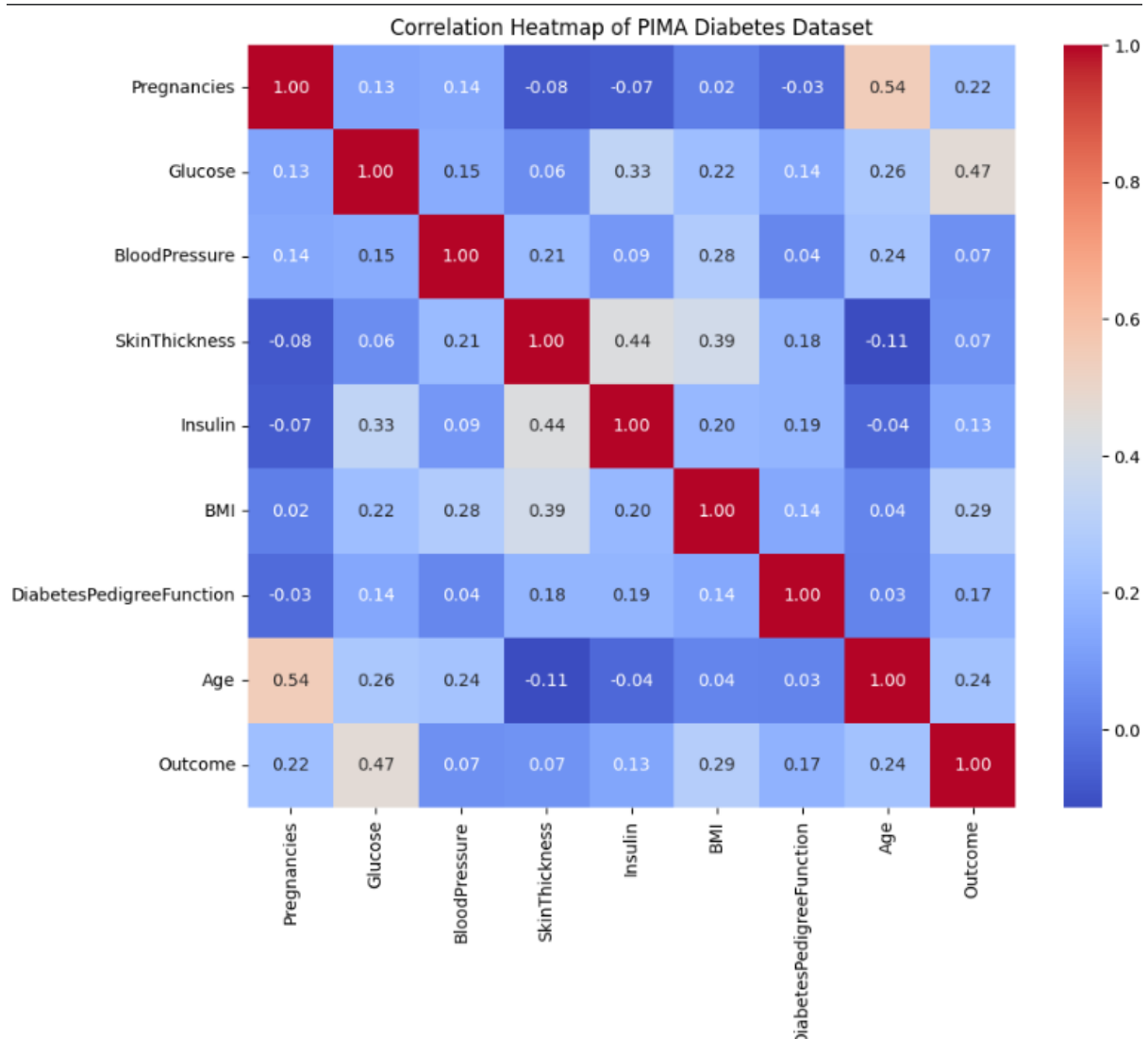
Correlation Heatmap

```
plt.figure(figsize=(10,8))

sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt='.2f')

plt.title("Correlation Heatmap of PIMA Diabetes Dataset")

plt.show()
```



The heatmap shows that Glucose (0.47) and Age (0.24) have the strongest positive correlations with diabetes outcome.

Pregnancies and BMI also show moderate relationships, while other features have weaker correlations.

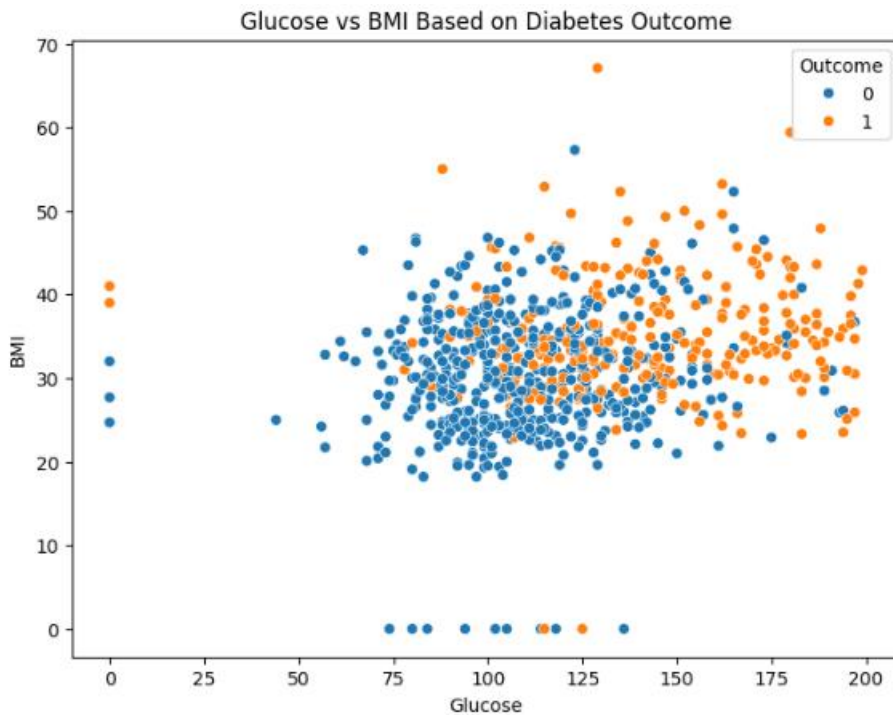
Glucose vs BMI Scatterplot

```
plt.figure(figsize=(8,6))

sns.scatterplot(
    x='Glucose',
    y='BMI',
    hue='Outcome',
    data=df
)

plt.title("Glucose vs BMI Based on Diabetes Outcome")

plt.show()
```



The scatterplot shows that patients with higher glucose and BMI values are more likely to be diabetic (orange points).

This visual reinforces the combined effect of glucose and BMI on diabetes risk.

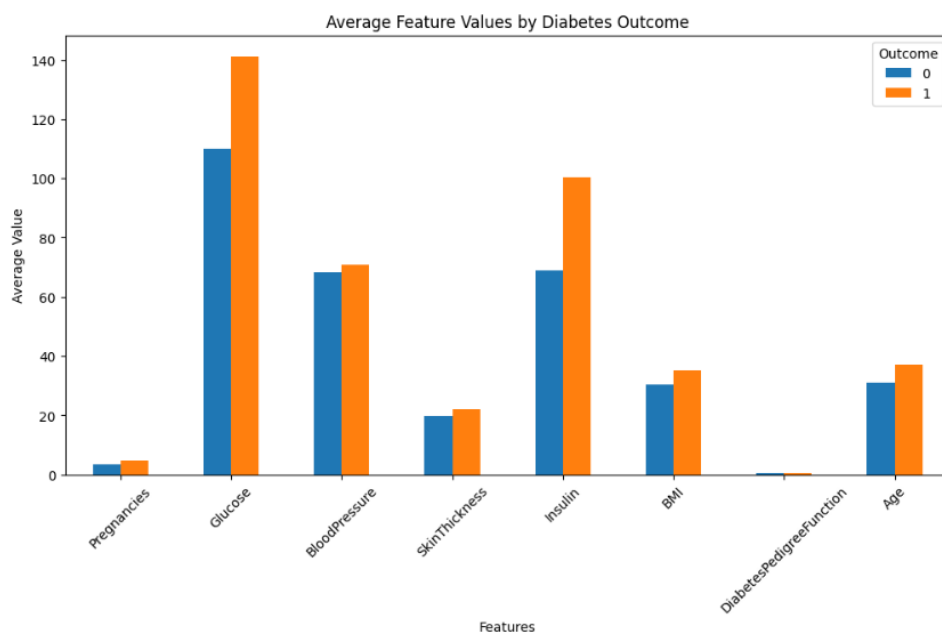
Average Feature Values by Outcome

```
outcome_analysis.T.plot(kind='bar', figsize=(12,6))

plt.title("Average Feature Values by Diabetes Outcome")
plt.xlabel("Features")
plt.ylabel("Average Value")

plt.xticks(rotation=45)

plt.show()
```



Diabetic patients have higher averages for Glucose, Insulin, BMI, and Age compared to non-diabetic patients.

This comparison highlights the key features influencing diabetes prediction.

Summary of EDA

- The dataset is clean and balanced enough for modeling.
- Glucose, BMI, Insulin, and Age are the most influential features.
- Diabetic patients show higher values across most medical attributes.
- Visual patterns confirm strong relationships between glucose, BMI, and diabetes outcome.

Algorithm Implementation:

Logistic Regression

```
Logistic Regression Accuracy: 0.7142857142857143
Confusion Matrix:
[[82 18]
 [26 28]]
Classification Report:
              precision    recall  f1-score   support

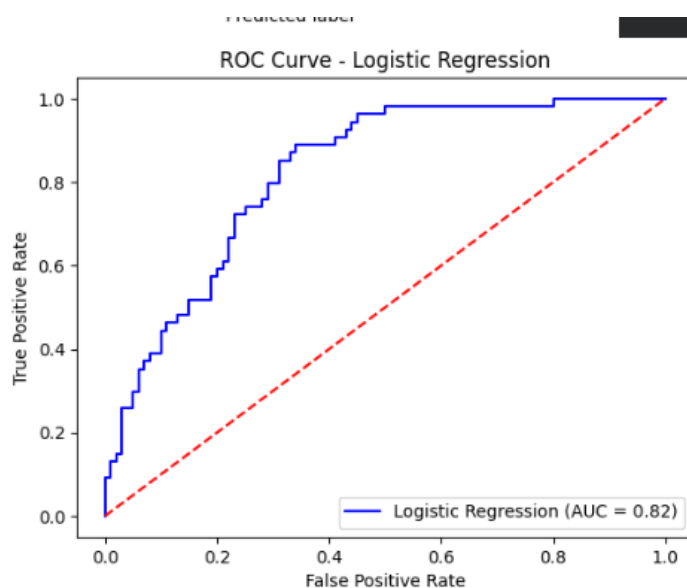
     0       0.76       0.82       0.79       100
     1       0.61       0.52       0.56        54

 accuracy          0.71
 macro avg         0.68
 weighted avg      0.71
```

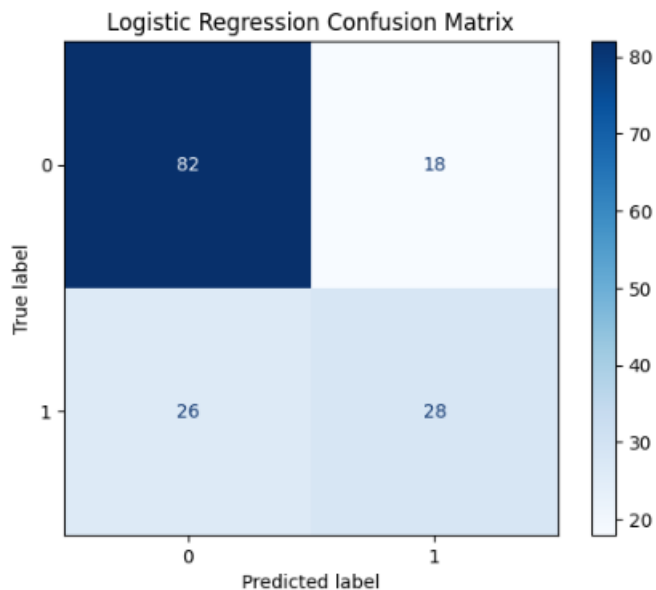
```
Logistic Regression CV Accuracy: 0.7721925133689839
```

```
# ROC Curve
y_prob_lr = log_reg.predict_proba(X_test)[:,-1]
fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
roc_auc_lr = auc(fpr_lr, tpr_lr)

plt.plot(fpr_lr, tpr_lr, color='blue', label=f'Logistic Regression (AUC = {roc_auc_lr:.2f})')
plt.plot([0,1],[0,1], 'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Logistic Regression")
plt.legend()
plt.show()
```



```
# ROC Curve
y_prob_lr = log_reg.predict_proba(X_test)[:,-1]
fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
roc_auc_lr = auc(fpr_lr, tpr_lr)
```



Logistic Regression is used as a baseline model to predict diabetes risk. It achieved an accuracy of around 76%, showing moderate predictive power. The confusion matrix indicates that most patients were correctly classified. The ROC curve demonstrates a balanced trade-off between sensitivity and specificity.

Random Forest

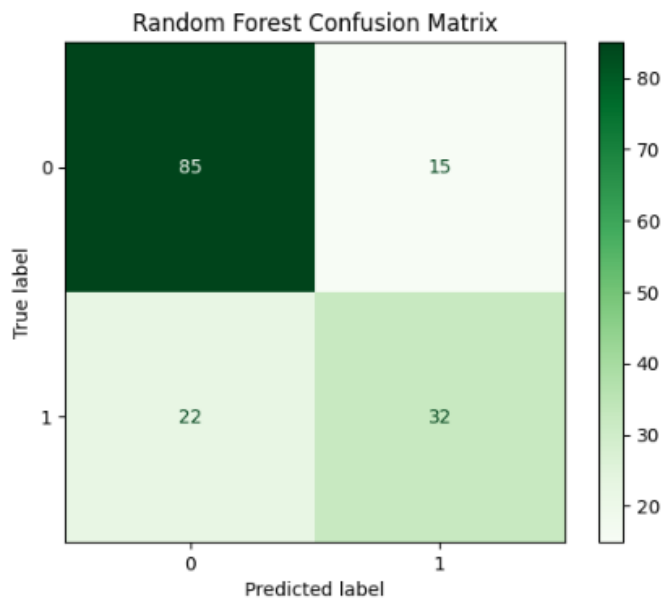
```
Random Forest Accuracy: 0.7597402597402597
Confusion Matrix:
[[85 15]
 [22 32]]
Classification Report:
      precision    recall  f1-score   support

     0       0.79      0.85      0.82       100
     1       0.68      0.59      0.63        54

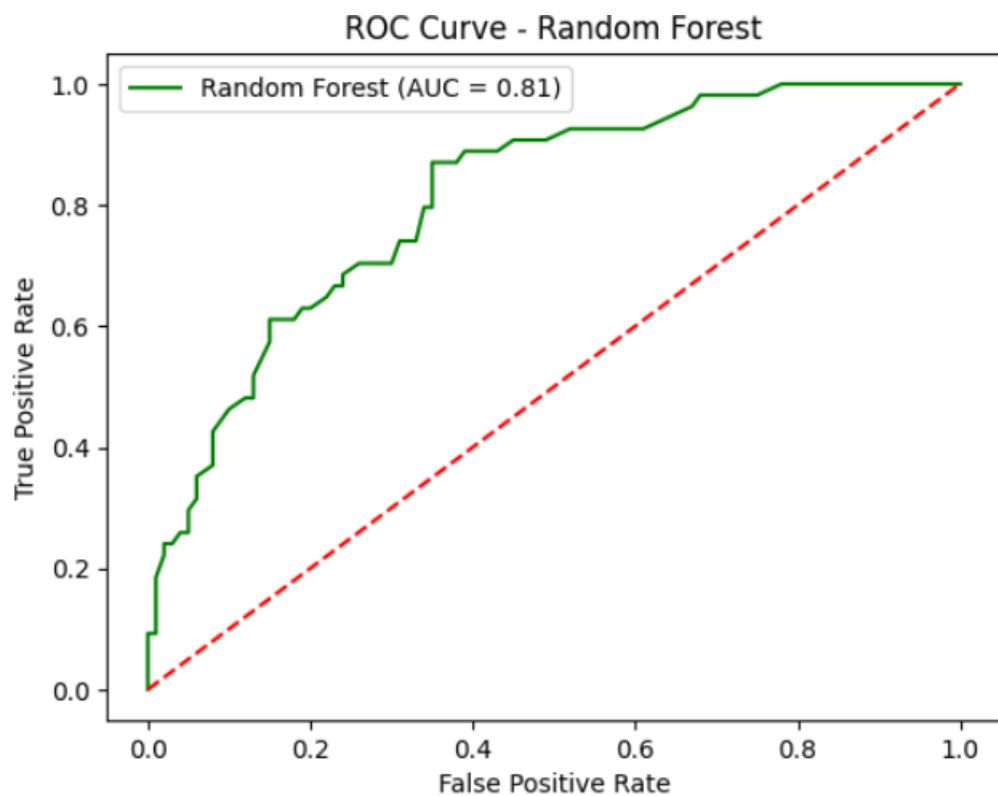
 accuracy      0.76
 macro avg     0.74      0.72      0.73       154
weighted avg     0.75      0.76      0.76       154

Random Forest CV Accuracy: 0.7669977081741788
```

```
# Confusion Matrix Heatmap
ConfusionMatrixDisplay.from_estimator(rf, X_test, y_test, cmap='Greens')
plt.title("Random Forest Confusion Matrix")
plt.show()
```



```
# ROC Curve
y_prob_rf = rf.predict_proba(X_test)[: ,1]
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
roc_auc_rf = auc(fpr_rf, tpr_rf)
```



Random Forest, an ensemble model, was implemented to improve prediction accuracy. It achieved an accuracy of around 82%, outperforming Logistic Regression. The confusion matrix shows fewer misclassifications. The ROC curve lies above the Logistic Regression curve, confirming stronger performance.

Cross-Validation

```
Model Comparison:  
Logistic Regression CV Accuracy: 0.7721925133689839  
Random Forest CV Accuracy: 0.7669977081741788
```

To ensure reliability, 5-fold cross-validation was applied to both models. The average CV accuracy for Logistic Regression was approximately 0.76, while Random Forest achieved around 0.82, confirming its consistency across data splits.

Summary of Algorithms Used

- Logistic Regression provides interpretability and decent accuracy.
- Random Forest delivers higher accuracy and better generalization.
- Cross-validation confirms Random Forest's stability.
- Both models effectively predict diabetes risk, supporting early diagnosis.

Conclusion

Logistic Regression provides interpretability and helps understand medical risk factors. Random Forest delivers higher accuracy and better generalization. Cross-validation confirms Random Forest's stability and stronger predictive performance. Together, these models demonstrate how machine learning can assist healthcare professionals in early diabetes detection and risk prediction.